

WASP-12b

R-1663

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_191025 · created 2026-08-02T09:43:05+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458781.88351 +/- 0.00068 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.096 +/- 0.045
Transit depth (Rp/Rs)^2	0.93 +/- 0.86 [%]
Orbital Inclination (inc)	76.85 +/- 2.84
Airmass coefficient 1 (a1)	0.8 +/- 0.46
Airmass coefficient 2 (a2)	0.26 +/- 0.53
Scatter in the residuals of the lightcurve fit is	60.09 %
Best Comparison Star	#3 - [150, 103]
Optimal Aperture	0.75
Optimal Annulus	3.75
Transit Duration (day)	0.099 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.096 $\pm$ 0.045 vs 0.1178 (deviation 0.35 σ)
Duration fit vs published	0.099 d vs 0.125 d (deviation 1.11 σ)
Mid-time O-C	-1.7 min
Depth SNR	1.6
Residual scatter	60.09 %

Light curve

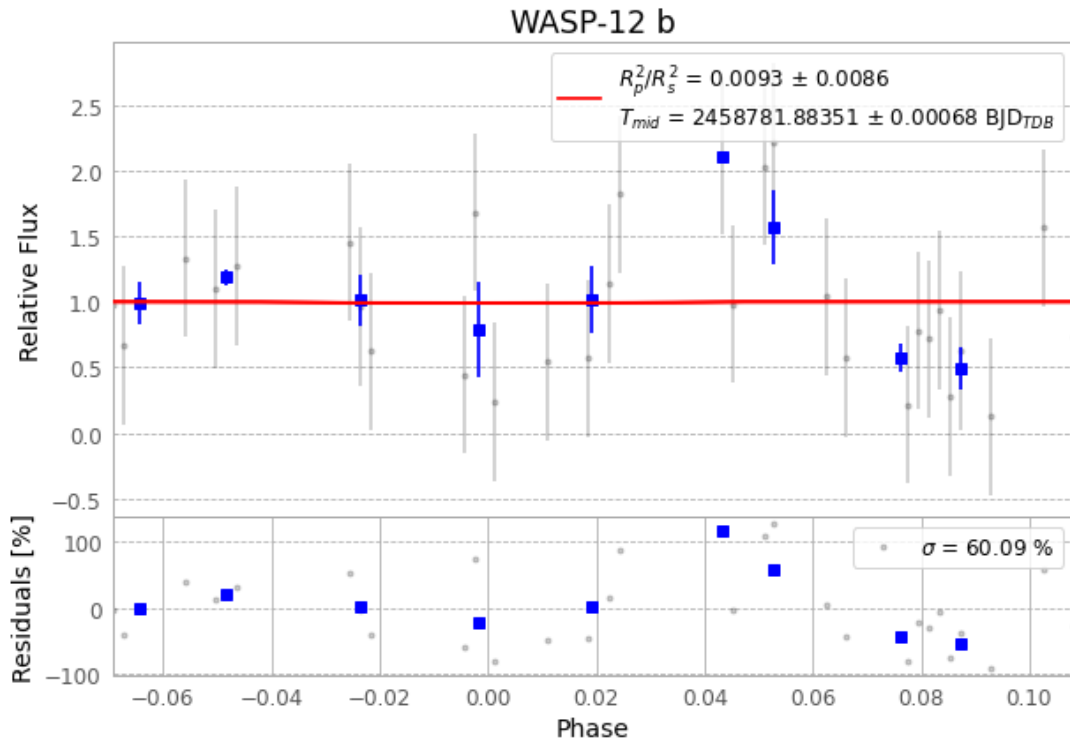


Figure 1 — FinalLightCurve_WASP-12 b_2019-10-25.png (SHA-256 d4a9eabd0bcb9874...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [308, 209], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 cde23cfefc9b4537...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

98 FITS frames (64 MB), WASP-12191025070612.FITS → WASP-12191025115712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-191025.log (a0234ecc2b75...), FinalLightCurve_WASP-12 b_2019-10-25.png (d4a9eabd0bcb...), FinalLightCurve_WASP-12 b_2019-10-25.csv (ed95d8e47d0f...)

Compute resources

Wall time 145.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 09:43Z.